Send Signal – Product Requirements Document

Executive Summary

Send Signal is a web-based application that empowers companies to automatically send **personalized WhatsApp messages** to new leads acquired from social marketing platforms (like Twitter and TikTok). The platform bridges the gap between social media lead generation and real-time personal engagement by using message templates with dynamic placeholders (e.g., “Hello **{Name}**”) to tailor each outreach. Send Signal aims to dramatically increase lead response and conversion rates.

Problem Statement

Modern marketing campaigns on platforms like Twitter and TikTok often generate many leads (with contact info such as phone numbers), but converting those leads into customers is challenging. Teams typically resort to **manual follow-ups** or impersonal bulk emails that yield low engagement. Key problems include:

- **Low Engagement via Traditional Channels:** Emails and generic SMS may be ignored or marked as spam. But, WhatsApp messages have higher visibility and read rates, but manually messaging each lead on WhatsApp is time-consuming and error-prone.

- **Lack of Personal Touch at Scale:** While personalization greatly improves response (e.g “Hi Mary, thanks for signing up!”), it’s difficult to personalize messages in bulk without a tool. Companies need a way to inject personal details into messages automatically so that each communication feels one-to-one.
- **Fragmented Lead Management:** Leads captured from Twitter, TikTok, and other platforms often live in spreadsheets. Without integration, staff must copy-paste numbers and craft individual WhatsApp messages, leading to delays and inconsistent follow-ups.
- **Compliance & Consistency:** WhatsApp has strict rules (opt-ins, template approvals) and without a proper system, businesses risk violating policies or sending inconsistent messaging. There’s a need for a compliant, guided process for WhatsApp outreach to ensure every message follows WhatsApp’s business guidelines (e.g., using approved templates with placeholders).

In summary, companies are losing potential sales due to slow, impersonal follow-ups. Send Signal addresses this by automating personalized WhatsApp communication at scale, ensuring leads are engaged instantly with tailored messages in a compliant manner.

Product Vision and Goals

Vision: To become the go-to platform for lead engagement, turning cold leads into warm conversations through personalized WhatsApp messaging. Send Signal envisions a world where businesses of any size can achieve *personal, high-touch communication at scale* as if they had a thousand hands texting each customer "personally," yet effortlessly automated.

Core Goals:

- **Instant Personal Connection:** Enable businesses to initiate a WhatsApp conversation with a new lead within minutes of capture, using the lead's name and context so it feels like a one-on-one message
- **Ease of Use:** Provide a **no-code, user-friendly interface** where marketing or sales folks (even non-technical) can create WhatsApp campaigns with personalization tokens (e.g., a placeholder for first name) and schedule or send them in a few clicks.
- **Compliance and Trust:** Ensure all outbound WhatsApp messages follow Meta's guidelines (using the WhatsApp Business API). This means handling opt-in verification, template message formatting, and unsubscribe handling gracefully so that businesses maintain trust and avoid WhatsApp bans.
- **Scalable Engagement:** Even as a company's lead volume grows, Send Signal should scale effortlessly. sending thousands of personalized messages reliably, tracking responses, and allowing team collaboration. The experience should scale from a small startup with 50 leads to an enterprise with 50,000 leads *without breaking the personal touch*.

Target Users and Use Cases

Primary Target Users:

- **Digital Marketers & Growth Hackers:** Individuals or teams running campaigns on TikTok, Twitter, Instagram, etc., who collect leads (via lead generation ads or forms) and need to convert them quickly.
- **Sales Development Representatives (SDRs) & CRM Managers:** Those responsible for follow-ups on inbound leads. They can use Send Signal to automate the initial touchpoint (e.g., a welcome message or product introduction) while preserving a human tone. This frees them to focus on hot leads who respond.
- **Small Business Owners & Agencies:** Small business owners who advertise on social media and lack dedicated staff for manual follow-ups. Agencies managing lead generation for multiple clients also benefit, using Send Signal to handle outreach for each client's leads in a centralized tool.

Key Use Cases:

- *Use Case 1 – Immediate Welcome Message:* A fitness studio runs TikTok ads with a free trial offer. As soon as a prospect submits their number, a message is sent from Send Signal; **"Hello [Name], thanks for your interest!"**, personalized with their first name and class of interest. The prompt, warm greeting makes the lead feel valued and more likely to convert.
- *Use Case 2 – Personal Lead Nurturing:* A B2B software company hosts a webinar promoted on Twitter. After the event, 100 attendees' contacts are funneled into Send Signal. The marketing manager schedules a follow-up WhatsApp campaign: "Hi [Name], thanks for attending our webinar. Let's chat about how [Company] can help [Lead's Company]." Each message auto-inserts the person's name and company, creating a customized touch

- *Use Case 3 – Cart Recovery via WhatsApp:* An e-commerce brand collects WhatsApp leads via a Twitter promotion. If a lead browses products but doesn't purchase, the team can send a **personalized WhatsApp message** (e.g., "Hi [Name], we noticed you liked [Product]. Here's a 10% off code just for you!"). This real-time, personal outreach can reclaim potentially lost sales (similar to how top WhatsApp tools enable cart recovery messages).

These use cases illustrate how Send Signal serves anyone who needs to **quickly turn a list of new leads into personalized WhatsApp conversations**. The platform is broad enough for creative use (sales, marketing, support follow-ups) but focused on leveraging WhatsApp to boost engagement.

Key Features (MVP)

The Minimum Viable Product will focus on core features that enable sending personalized WhatsApp campaigns to imported leads, with simplicity and compliance in mind:

- **Lead Import & Integration:** For MVP, this includes:
    - **CSV Upload & Manual Entry:** Users can upload a CSV of leads (with columns like Name, Phone, etc.) or add a single lead manually.
- **Contact Management:** A simple Leads dashboard to view and manage contacts. Users can see each lead's details (name, phone, source, timestamp acquired) and their engagement status (e.g., message sent, replied, opted-out). Basic filtering by source or tags is included so users can segment (e.g., separate TikTok leads from Twitter leads).
- **Personalized Message Templates:** The ability to create, save, and reuse message templates with placeholders. Users can type a message and insert variables for personal details, such as **{FirstName}**, which get replaced per

lead. For example: *"Hi {FirstName}, thanks for signing up via {Source}!"* would become "Hi *Jane*, thanks for signing up via *TikTok*!" on send. This ensures at send-time each lead's message is unique and personal.

- **WhatsApp Messaging (Bulk Send):** Integration with the WhatsApp Business API (using an official provider) to send out messages.
- **Opt-In Management & Compliance:** Built-in compliance features so that only leads who have consented to WhatsApp contact are messaged.
- **Analytics Dashboard:** A simple analytics panel showing key metrics:
    - Number of leads imported (by source).
    - Campaign performance: send counts, delivery rate etc. For example, *"WhatsApp messages sent: 1000, Delivered: 980 (98%), Read: 900 (90%), Replies: 150 (15%)"* – highlighting WhatsApp's high engagement.
    - Conversion tracking (optional/basic): Users can manually mark a lead as "Converted" or input if that lead made a purchase, etc., to see how messaging is impacting conversions.

User Account & Settings:

- **WhatsApp Business Account Connection:** The user will need to register their WhatsApp Business number through that provider and connect API keys in Send Signal.
- **Team Access:** MVP may support a single user per account (one login). If simple, we could include inviting team members to collaborate.

Future Features and Roadmap

As Send Signal grows, we plan to expand capabilities to provide even more powerful engagement and automation. Future features on the roadmap include:

- **Advanced Automation & Drip Campaigns:** Ability to set up multi-step messaging sequences on WhatsApp. For example, *Day 1:* send welcome message, *Day 3:* send a follow-up tip or product highlight, *Day 7:* send a reminder or promotional offer – all automated once the lead enters a campaign. Users could configure triggers (e.g., stop sequence if lead replies or converts). This turns Send Signal into a full WhatsApp drip marketing tool, similar to email automation but over chat.
- **Chatbot Integration:** Integrate or provide a no-code chatbot builder to handle common lead replies automatically. E.g., if a lead responds with “Tell me more about pricing,” a bot could automatically reply with pre-set info. This would leverage AI or rule-based logic to simulate a conversational assistant, enhancing scalability of two-way communication (and aligning with trends where platforms offer AI chat on WhatsApp).
- **CRM Integration & Sync:** Deep integration with popular CRMs (HubSpot, Salesforce, Zoho, etc.). This means leads’ status and conversations on Send Signal sync back to the CRM, and vice versa. A sales team working in CRM could see WhatsApp conversation history, and new contacts added in CRM could be auto-imported to Send Signal for messaging. This bi-directional sync ensures Send Signal fits seamlessly into existing sales pipelines.

Free vs Paid Feature Tiers

Send Signal will adopt a **freemium model** to encourage widespread adoption, with a robust free tier and value-added paid plans for power users. Below is an outline of the tiers and their features:

Free Tier (Launch Plan – "Starter"):
Designed for individual marketers or very small businesses to try out Send Signal. Includes:

- **Core Messaging Features:** Ability to send a limited number of WhatsApp messages per month (e.g., *up to 100 messages* free) with personalization. This allows testing of the service with real leads.
- **Basic Integrations:** One-click CSV import and manual lead entry. Possibly connect one lead source (choose either Twitter OR TikTok integration in free tier).
- **Single User:** Only one user login (the account owner) can access the dashboard.
- **Basic Support:** Access to email support and self-serve knowledge base. Response times may be slower for free users.

Paid Tier(s):
We'll offer graduated plans to cater to growing teams and larger needs.

Pro Plan: A monthly subscription (e.g., $20/month) aimed at small-to-medium businesses and agencies.

- **Higher Message Quota:** Send a larger volume of messages (e.g., *5,000 messages/month*, with affordable overage pricing per additional message beyond that).
- **Multiple Integrations:** Connect multiple lead sources (Twitter, TikTok, Facebook Lead Ads, etc. we'll expand integration options here).
- **Multi-User Access:** e.g., 5 team members can collaborate in the account. This includes shared inbox usage for reply handling.
- **Advanced Features:** Unlock features such as scheduling messages , drip campaigns, and A/B testing of two message variants.
- **Analytics Reports:** More detailed reporting and the ability to export data (e.g., CSV export of message logs).
- **Priority Support:** Faster response, possibly live chat support during business hours.

Enterprise Plan: Custom pricing (likely usage-based or negotiated) for large companies or agencies with massive volume.

- **Unlimited or Very High Limits:** Send tens of thousands of messages per month, with dedicated throughput (possibly using multiple WhatsApp Business API phone numbers to handle volume if needed).
- **All Integrations & APIs:** Access to all integrations and a **Send Signal API** so the enterprise can programmatically trigger WhatsApp messages or sync data from their systems.
- **Enhanced Security & Compliance:** Features like single sign-on (SSO), audit logs, GDPR compliance guarantees with data processing agreements, and on-demand data deletion for privacy requests.
- **Account Management & Support:** A dedicated account manager, 24/7 support, and onboarding assistance (including help with WhatsApp Business API approvals or number setup).

Detailed User Flows

Below are detailed step-by-step user flows for key activities in Send Signal.

User Onboarding & Account Setup

1. **Sign Up:** The user visits the Send Signal web app and signs up (provide email, set password, or use Google OAuth if available). After email verification, they're taken to an onboarding wizard.
2. **Connect WhatsApp API:** The wizard guides the user to connect a WhatsApp Business number. For MVP, this might show instructions or integrate with a provider:

    - If using WhatsApp Cloud API: user is prompted to log in to Facebook/Meta to authorize their WhatsApp Business Account (WABA) access.
    - Or the scan a Whatsapp QR Code

3. **Import/Connect Lead Source:** Next, the wizard asks how the user wants to bring in leads:
    - Option A: Upload CSV : user selects a CSV file from their computer, maps the columns (e.g., Column1 = Name, Column2 = Phone, etc.), and uploads. They see a preview of how many leads were added.

4. **First Template Creation:** The wizard might encourage the user to create their first message template. It could present a sample template (e.g., "Hi {Name}, thanks for joining from {Source}!") which the user can customize. The interface highlights where to insert placeholders (maybe a dropdown of available fields).

5. **Review Dashboard:** On finishing onboarding, the user lands on the main dashboard with a brief tour highlighting key sections: Leads, Templates, Campaigns, Inbox, etc. This completes the initial setup.

Sending a Personalized WhatsApp Campaign

1. **Initiate Campaign:** User clicks “New Campaign” (prominently on dashboard or Campaigns section). A wizard or form opens for sending a message.

2. **Select Audience:** Step 1 – user chooses recipients. Options might include:
    - *All Leads* (with maybe a count shown, e.g., “All 250 leads”),
    - *Segment/Filter* users can filter (for MVP, by source or tag or import batch). E.g., filter to “Source = TikTok” or manually check a subset of leads from a list.
3. **Compose or Choose Template:** Step 2 – user composes the message. They can either pick one of their pre-saved templates from a dropdown **or** type a new custom message. For example, they type “Hello ” then click **Personalize -> First Name**, and it inserts the token. A live preview on the right might show an example substitution (e.g., preview with the first lead: “Hello John, …”).
4. **Review & Send:** Step 3 – user reviews summary: e.g., “Recipients: 120 leads, Message template: X (placeholders will be filled).”
    - The user chooses to either **Send Now** or **Schedule** for later
5. **Confirmation:** User hits send, gets a confirmation modal: “Your campaign is being sent to 120 leads. This may take a few minutes. You can monitor progress in the Campaigns dashboard.” They click confirm, and the campaign is queued.
6. **Sending Process (background):** The system starts sending messages, e.g., 20 per second to avoid throttling . The user can navigate to Campaigns list to see this campaign’s status (e.g., “Sending… 45/120 sent”).

7. **Result Monitoring:** After sending, the user sees final stats: out of 120, how many were sent successfully, how many failed (with reasons if possible, e.g., invalid number). The Campaign detail page and overall Analytics reflect these numbers.

Technical Requirements

Architecture Overview

The system will follow a **client-server web application architecture** with clear separation of concerns:

- **Frontend:** A web client (SPA : Single Page Application) built with a modern framework (e.g., React or Vue). This will handle the user interface, display of data, and user interactions. The frontend should be responsive (usable on various screen sizes, though primary use is desktop web) and optimized for performance for data-heavy views (like tables of leads).

- **Server-Side:** A server-side application (e.g., Node.js with Express, or Python with Django/Flask, etc.) that provides RESTful API endpoints for all core functionalities (lead management, sending messages, etc.). The backend handles business logic, communicates with external APIs
    - *Lead Service:* for storing and retrieving leads.

    - *Messaging Service:* to queue and dispatch WhatsApp messages (through WhatsApp API).

    - *Integration Service:* handles incoming data from webhooks (e.g., Twitter lead gen webhook or WhatsApp incoming messages).

- **Database:** A reliable database to store persistent data:
    - Likely a relational DB (PostgreSQL or MySQL) to store structured data: user accounts, leads, templates, messages log, etc. Relational makes sense for queries like join user->leads or filter leads by source.
    - Additionally, maybe a NoSQL store (like Redis) for caching frequently accessed data or ephemeral data (session store, rate limiting counters, etc.).
    - Message logs could become large; we might use a combination: store summary stats in DB, but detailed logs in a scalable log system if needed (or prune older logs).

Security & Compliance

Given we handle personal data (names, phone numbers) and integrate with various services, security is paramount:

- **Data Protection:** All data in transit will be over HTTPS (SSL encryption). For data at rest:
    - Encrypt sensitive fields in the database (especially WhatsApp auth tokens, API keys for integrations, perhaps phone numbers if required by policy – though phone numbers are used often so maybe not encrypted but definitely protected).

    - Regular backups of the database with secure storage (and encryption) to prevent data loss.

- **Authentication & Authorization:** Implement secure user auth. Possibly JWT-based sessions or server-side sessions with secure cookies. Ensure password storage is hashed (bcrypt/scrypt) if not using OAuth. Multi-factor auth could be a future addition for extra security.

    - Authorization checks on every API (a user can only access their leads, etc.). If multi-tenant architecture, isolate data properly by user/org.

- **Compliance with WhatsApp Policies:**

    - The system *must* enforce obtaining user consent (opt-in) before messaging. We might include a feature for users to import proof of consent or have a checkbox “I confirm these leads have opted into WhatsApp communication.” This is more on the business user, but we should remind and log that confirmation to protect us.

    - Must use **approved templates** for outbound messages. This means technically, we can’t just send arbitrary text to a new contact; it has to match an approved template ID via WhatsApp API. Our template mechanism and approval flow ensure compliance here.

    - Provide an easy opt-out as described: if someone says “STOP”, the user must honor that. We can automate adding them to a do-not-contact list on our end.

Metrics of Success

To gauge Send Signal's performance and success, we will track a variety of **key metrics**. These metrics help us understand user adoption, engagement, the value delivered to customers, and overall business health. Below are the crucial metrics categorized by aspect:

Product Adoption & Engagement Metrics:

- **Number of Active Users (DAU/MAU):** The count of daily active users or monthly active users on the platform.
- **Onboarding Completion Rate:** What percentage of sign-ups go through key onboarding steps (connecting WhatsApp, importing leads, sending first campaign).
- **Campaigns Sent per User:** Average number of campaigns or messages sent by an active user per week or month..
- **Feature Usage Metrics:** e.g., percentage of users using Template feature, or using Integrations vs manual upload. This helps identify which features are most valuable and which might need improvement or promotion.

Customer Success & Effectiveness Metrics:

- **Message Delivery and Read Rates:** Although partially dependent on WhatsApp and user's recipients, we track aggregate **delivery rate** (e.g., 98% delivered) and **read rate** (maybe 80% if most people read messages).

Business & Revenue Metrics:

- **Monthly Recurring Revenue (MRR) and Growth Rate:** Total subscription revenue monthly and its month-over-month growth.

- **Conversion Rate (Free to Paid):** What percentage of free tier users convert to a paid plan within a given period (e.g., within 1 month of sign-up).

- **Churn Rate:** The percentage of paying users who cancel or downgrade their subscription in a given month. Low churn is crucial for SaaS health.

- **Enterprise Pipeline:** For enterprise deals (which might be fewer but larger), track number of enterprise leads, deals closed, and average contract value. Even if these are not high volume, one enterprise client could boost revenue significantly, so we track them individually.

System Performance Metrics:

- **Uptime Percentage:** We aim for 99.9% uptime or better for the service. We monitor downtime incidents and resolution time. A stable platform retains users better.

- **Message Throughput & Latency:** Ensure our system can send messages promptly. E.g., time from campaign start to all messages sent. If this remains low (a few minutes for a large campaign), it's good. If users notice slowness, we might see it in feedback or support tickets, which also is something to track (# of support tickets or issues raised).

User Satisfaction Metrics:

- **CSAT (Customer Satisfaction) for support:** If we have support interactions, gather ratings. Quick and helpful support will keep users happy especially when dealing with issues around messaging or setup.

- **User Feedback & Reviews:** Track qualitative feedback, but also if we list the app on marketplaces or get reviews on sites, the average rating is a metric (e.g., 4.5/5).

Risks and Limitations

No project is without risks and constraints. Below, we outline potential risks for Send Signal along with limitations we need to manage, and in some cases mitigation strategies:

- **WhatsApp Policy Changes / Dependency Risk:** Send Signal heavily relies on WhatsApp's Business Platform. If WhatsApp (Meta) changes policies, pricing, or limits (e.g., drastically raising message costs, or restricting template types), our service could be impacted.

- **Message Deliverability and Spam:** If users misuse Send Signal to send spammy content or contact people without proper consent, WhatsApp might **block the number or ban the account**. This not only affects that user but could tarnish our IP/domain reputation if messages funnel through our service.

- **Competitive Risk:** The market for WhatsApp messaging and CRM tools is active. Established competitors could introduce similar personalization features or even undercut on pricing. We also face the risk of big CRMs (Salesforce, HubSpot) building more native WhatsApp integration that diminishes the need for a standalone tool.

- **Scalability Challenges:** If our user base grows quickly , our system might face scaling issues, slowdowns, message queue backups, etc. Additionally, sending large volumes of WhatsApp messages might hit provider limits (WhatsApp has tiered throughput based on quality rating).

- **User Adoption and Behavior Risk:** There is a risk that some target users may be hesitant to use WhatsApp for outreach, fearing it could be seen as intrusive.

- **Legal and Compliance Risk:** Aside from WhatsApp policies, we must consider laws like anti-spam regulations. Also storing EU user data likely means GDPR obligations (as noted).

- **Financial Risk:** As a startup product, if revenue doesn't ramp up as expected, there's risk around sustaining operations (paying for servers, etc.). Also unforeseen costs, like if WhatsApp messaging is more expensive due to new pricing, could hit margins.

- **Limitations in MVP:** The initial product won't do everything. Limitations include:

    - It's web-only (no mobile app yet), so users who want mobile notifications must rely on email notifications. Some might find that a limitation.

    - We handle WhatsApp text and maybe images, but not voice notes or other advanced WhatsApp features in MVP.

    - The personalization is basic (we can't magically know more about the lead than provided; if the lead info is sparse, messages are limited).

    - If a user wanted to use it for something like *ongoing chat support*, our tool is not as full-featured as a dedicated support platform (no multi-shift routing, etc. in MVP). We focus on marketing/sales outreach use cases.

    - We rely on the user's WhatsApp Business API access; if a small user doesn't have that or struggles to get approved by Meta, it limits their use. We'll have to guide them or possibly offer a shared sandbox for small volume testing, but for production they need their number.

Competitive Landscape and Analysis

Send Signal operates in a space that overlaps personalized WhatsApp communication, marketing automation, and CRM-style engagement tools. Understanding the competitive landscape is crucial for identifying our unique value proposition. Below is a list of well-known global tools and platforms in this domain, along with a brief summary of each competitor:

CleverTap: A comprehensive customer engagement platform that is also an official WhatsApp Business Solution Provider. CleverTap emphasizes personalization at scale and omnichannel campaigns. It provides a no-code way to launch WhatsApp campaigns with rich segmentation and AI-driven optimization, turning WhatsApp into a channel for measurable growth

Respond.io: A multi-channel messaging platform that unifies conversations from various channels (WhatsApp, Messenger, Instagram, etc.) into one inbox. It features a drag-and-drop automation builder and robust analytics. *Summary:* Great for businesses that want a single hub for all messaging channels, not just WhatsApp.

Zoko: A user-friendly WhatsApp business platform geared towards e-commerce. Zoko consolidates customer chats, marketing automation, and order management on WhatsApp. It integrates deeply with Shopify to enable features like cart recovery messages and product catalogs in WhatsApp.

AiSensy: A WhatsApp marketing platform focusing on easy automation and no-code chatbots. AiSensy allows broadcasting messages, building chat flows without coding, and supports multi-agent chat handling. It also integrates with Shopify, HubSpot, etc.

Competitor Summary Table:

Platform	Strengths	Focus Area	Notes
CleverTap	Personalization, analytics, AI	Enterprise marketing (multi-channel)	BSP with deep segmentation and journeys.
WATI	Shared inbox, easy automation	SMB team WhatsApp support	Official BSP, templates and integrations.
Twilio API	Developer flexibility, scalable infra	WhatsApp API (no UI)	Requires coding; great for custom builds.
Respond.io	Multi-channel inbox, automation	Unified communications	Handles WhatsApp, FB, IG in one place.
Zoko	E-com integration (Shopify), UI simplicity	WhatsApp sales & support	Cart recovery and team inbox features.
AiSensy	No-code chatbots, broadcasts	WhatsApp marketing (India)	User-friendly, includes chatbot builder.